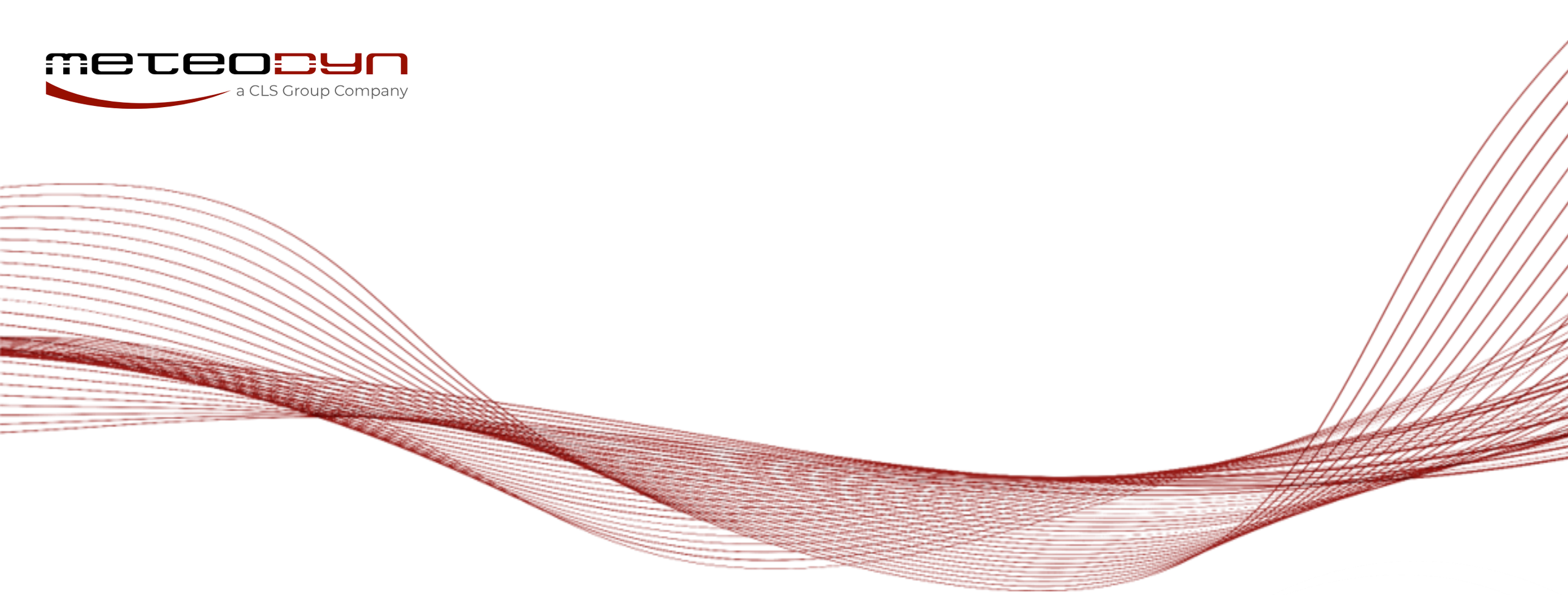
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# Wind Farm Performance Analysis Methodology and Cases

1 Performance analysis solutions for wind farms

2 WPA technical basis

3 Performance analysis cases

4 How to carry out wind farm performance analysis with WPA

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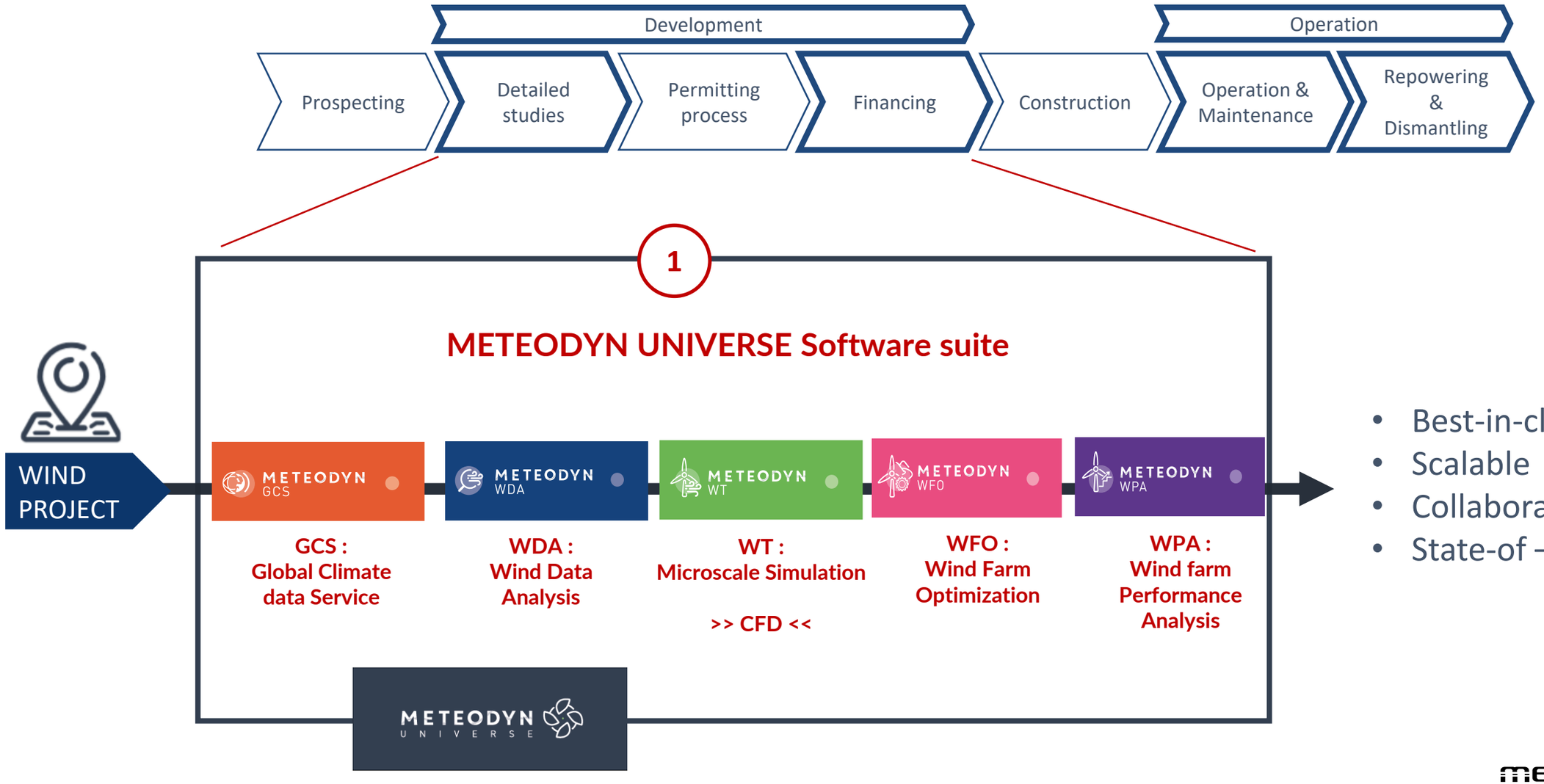
1 Performance analysis solutions for wind farms

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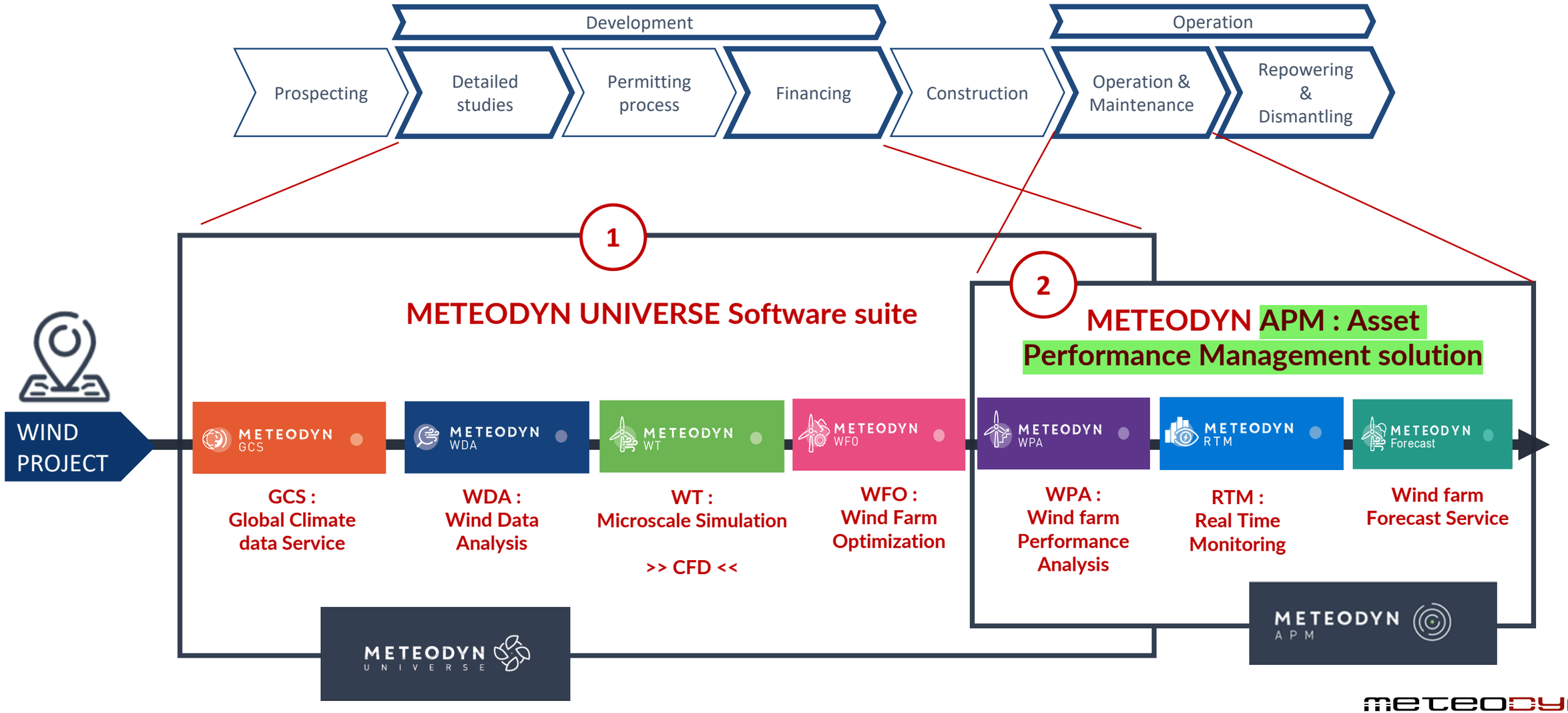
4 How to carry out wind farm performance analysis with WPA

# 1 UNIVERSE suite ➔ WT = CFD (Computational Fluid Dynamics)




- Best-in-class precision
- Scalable
- Collaborative
- State-of –the-art

# 2 APM software ➔ Real Time Monitoring



# WPA - Wind farm Performance Analysis

METEODYN WPA

## High compatibility

At the very least, only the time series of wind speed and power need to be provided for analysis, and no input of the operating status of the wind turbine (e.g., power rationing, fault) is required.

## Fast and intelligent analysis

Based on the intelligent algorithm developed by Meteodyn, it can quickly identify the operating status, and calculate the power generation loss.

## KPI calculations

Evaluate key performance indicators such as power curve, TBA, PBA, actual power generation, and power loss to help users efficiently compile energy efficiency analysis reports for wind farms.

## Comprehensive analysis & visualization

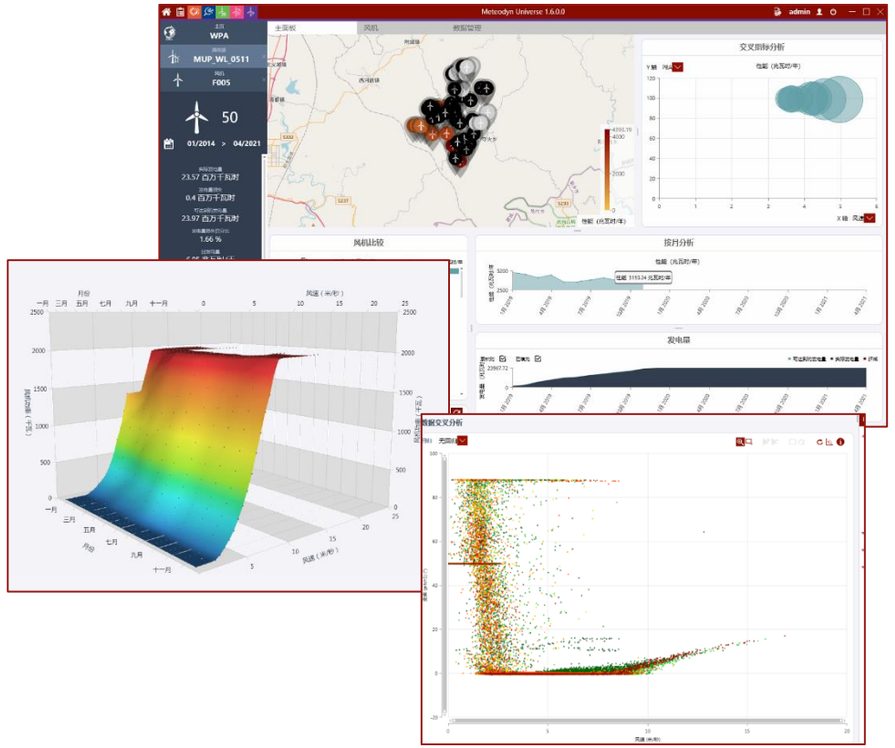
Provide detailed visual analysis charts and intuitive data cross-display analysis functions to help users quickly realize unit performance diagnosis and explore the value of SCADA data.

## Follow industry standards

The treatment is in accordance with IEC 61400-12-1, IEC 61400-12-2 and FGW TG10 technical guidelines.

## Capitalization & collaboration

Raw data and analyses are stored in a software database, allowing users to query past analyses at any time, add new data to continue their analysis, and share their work with other collaborators within the organization.




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1 Performance analysis solutions for wind farms

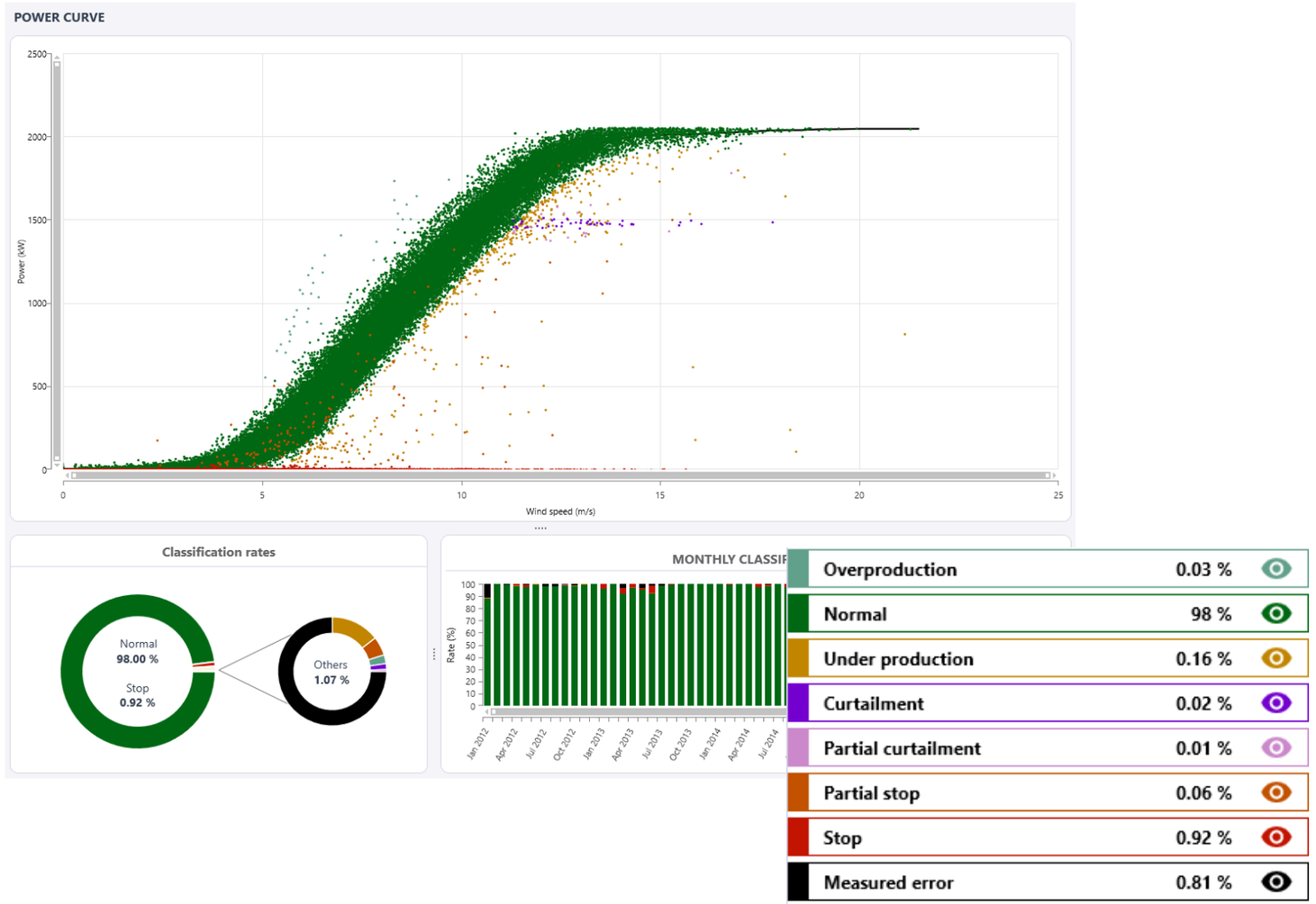
2 WPA technical basis

3 Performance analysis cases

4 How to carry out wind farm performance analysis with WPA

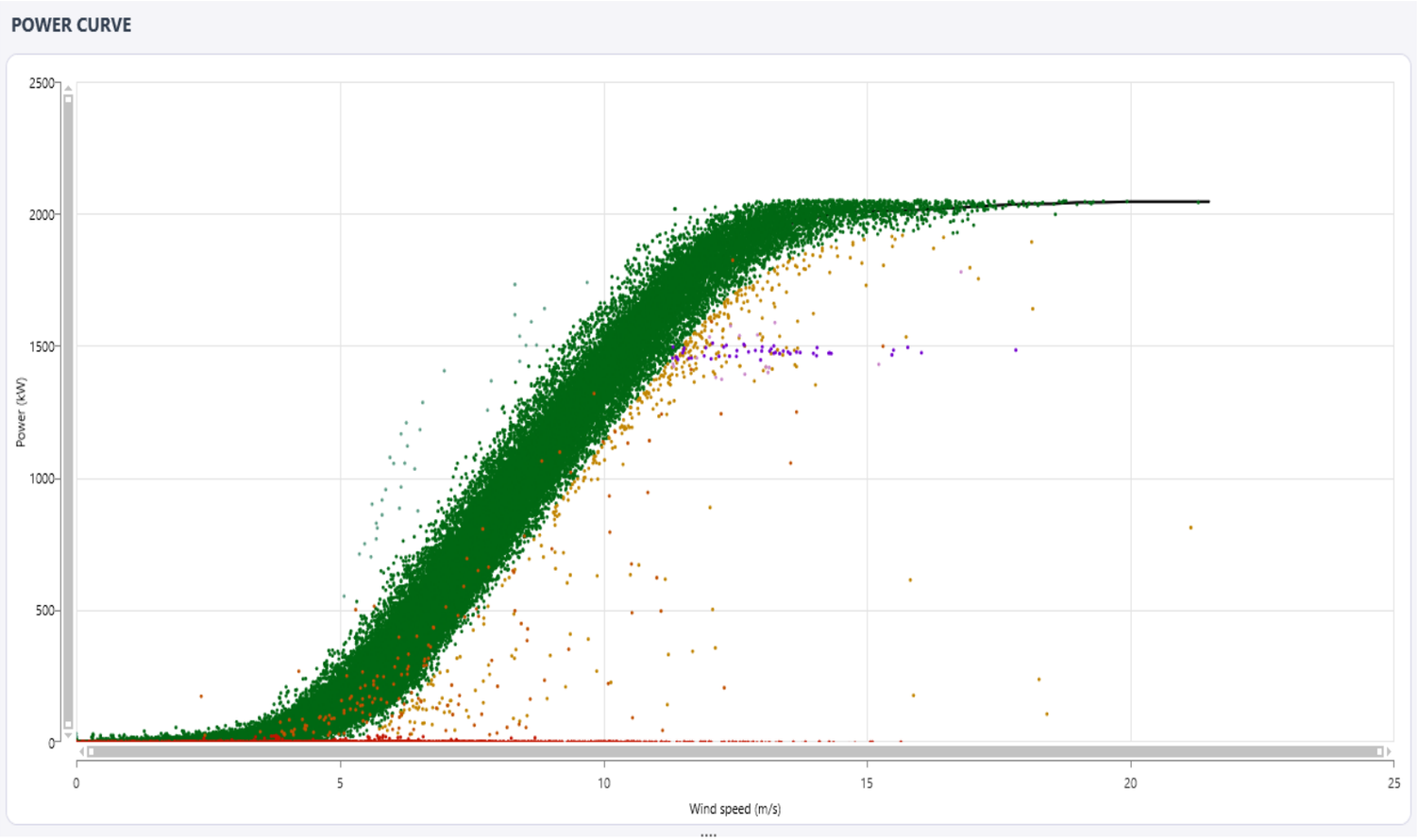
# Batch import of SCADA data and classification of operating status

- Quickly import historical SCADA data for all units in a wind farm
- Automatic data quality checking
- Based on the robust algorithm developed by Meteodyn, the operating status of the unit is intelligently identified and classified into 8 categories:
  - normal
  - stop
  - Under production
  - Over production
  - Curtailment
  - Partial stop
  - Partial curtailment
  - Measurement errors or missing measurements



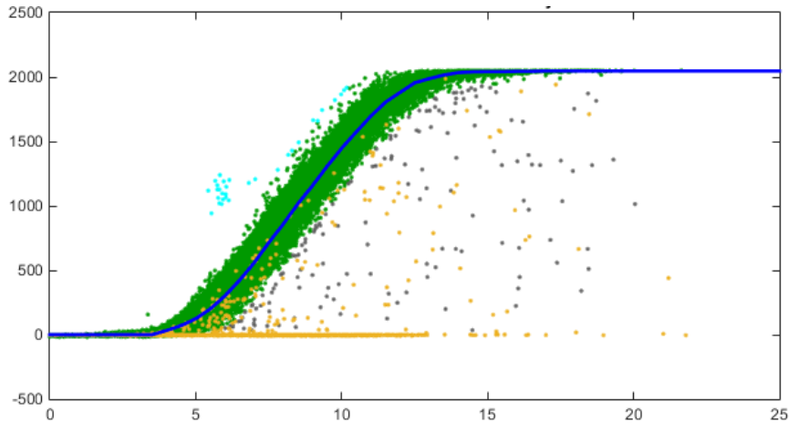
# Reconstruct the power curve

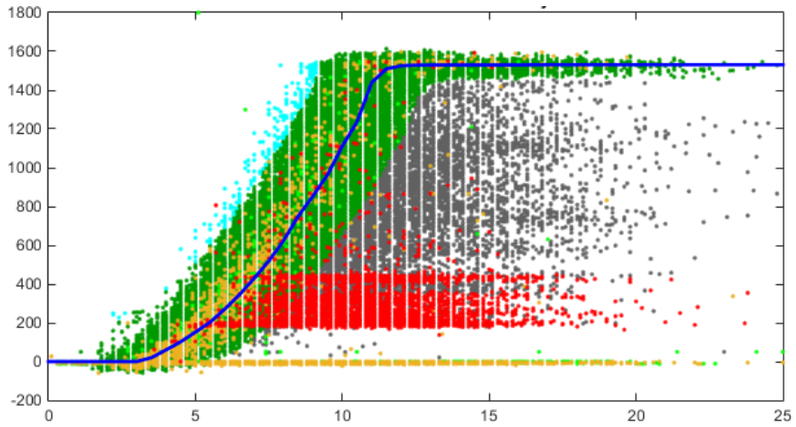
- The actual power curve is calculated based on the data of the normal operating state
- The power curve is calculated in accordance with IEC 61400-12-1

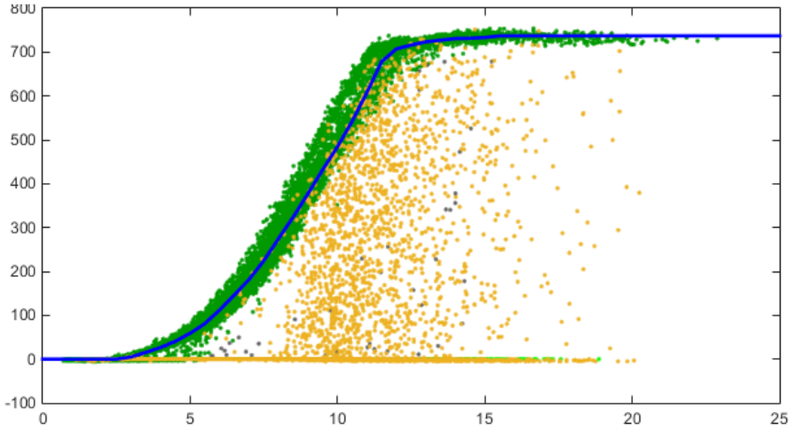


# Validation of the operational status classification method

Validation on a large number of real SCADA data shows that WPA's data classification algorithm is very robust and still valid even when the normal data proportion is less than 50%.



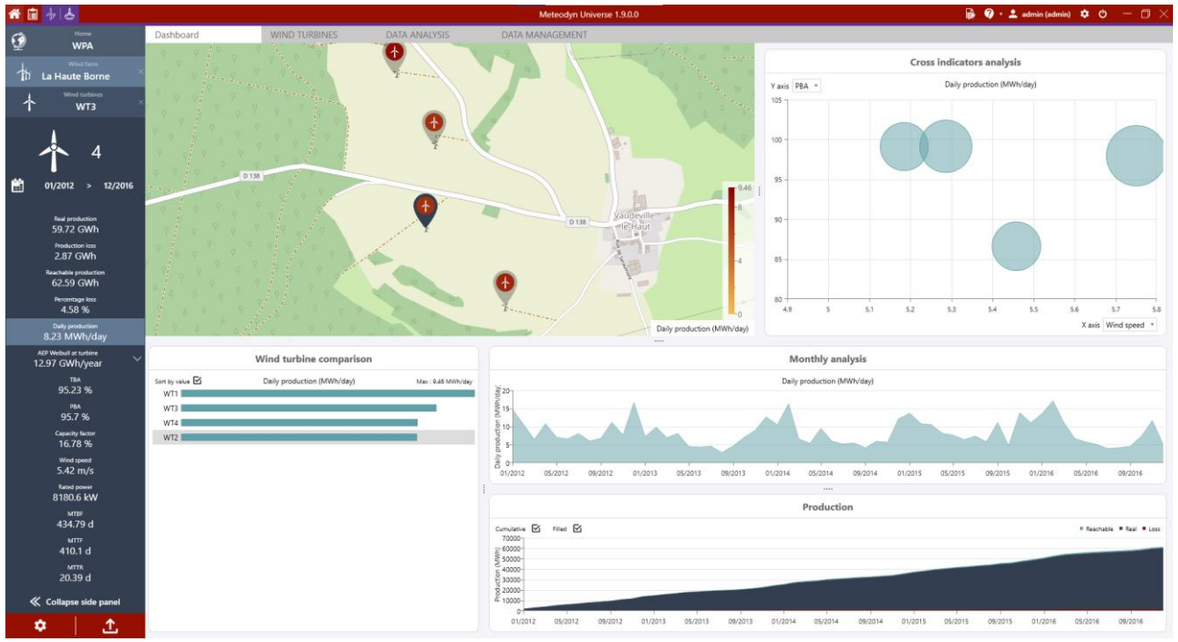



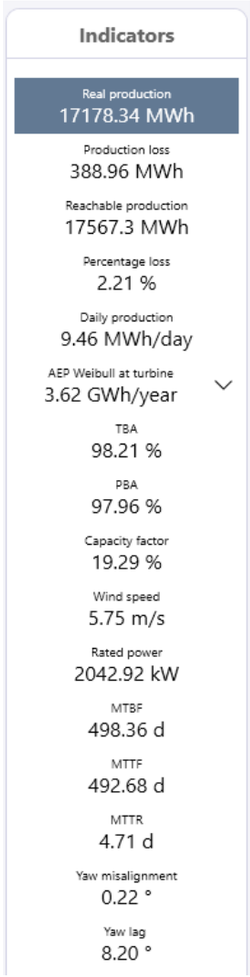




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# Calculate KPI for wind farms and wind turbines

- Automatic calculation of performance indicators for wind farms and wind turbines
- Quickly identify anomalies based on the comparison of key indicators
- Provide more than 20 performance indicators:
  - Actual production/reachable production
  - Loss due to different factors
  - Proportion of different running states
  - Availability (TBA/PBA)
  - AEP based on the Weibull distribution of the wind turbine
  - AEP based on the Weibull distribution of wind farm
  - AEP based on Rayleigh distribution
  - Fault metrics (MTBF, MTTR, MTTF)
  - ……



# Power curve correction

- Air density correction (IEC 61400-12-1)

$$\rho_{10\min} = \frac{1}{T_{10\min}}\left(\frac{B_{10\min}}{R_0} - \Phi P_w\left(\frac{1}{R_0} - \frac{1}{R_w}\right)\right)$$

where

| | |
|---|---|
| $\rho_{10\min}$ | is the derived 10 min averaged air density; |
| $T_{10\min}$ | is the measured absolute air temperature averaged over 10 min [K]; |
| $B_{10\min}$ | is the air pressure corrected to hub height averaged over 10 min [Pa]; |
| $R_0$ | is the gas constant of dry air 287,05 [J/kgK]; |
| $\Phi$ | is the relative humidity (range 0 % to 100 %); |
| $R_w$ | is the gas constant of water vapour 461,5 [J/kgK]; |
| $P_w$ | is the vapour pressure equal to 0,000 020 5 exp(0,0631 846 $T_{10\min}$) [Pa]; |

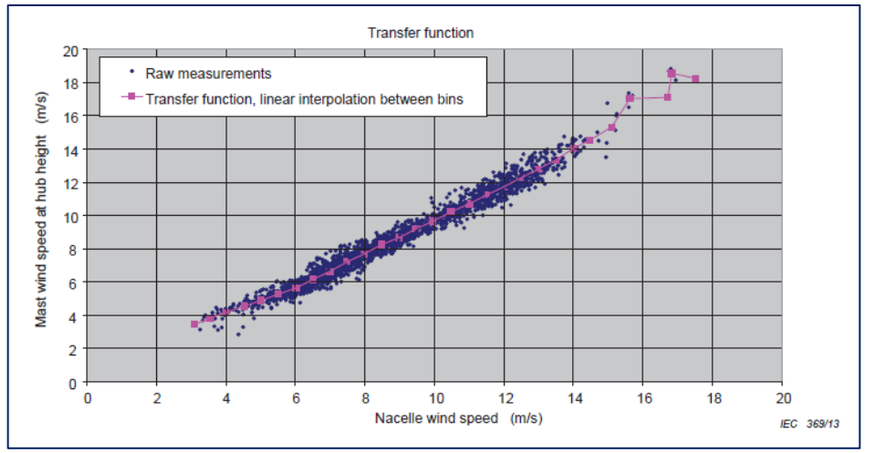
$$V_n = V_{10\min}\left(\frac{\rho_{10\min}}{\rho_0}\right)^{1/3}$$

where

| | |
|---|---|
| $V_n$ | is the normalised wind speed; |
| $V_{10\min}$ | is the measured wind speed averaged over 10 min. |

- NTF correction (IEC 61400-12-2)

$$V_{free} = \frac{V_{met,i+1} - V_{met,i}}{V_{nacelle,i+1} - V_{nacelle,i}}\left(V_{nacelle} - V_{nacelle,i}\right) + V_{met,i}$$



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1 Performance analysis solutions for wind farms

2 WPA technical basis

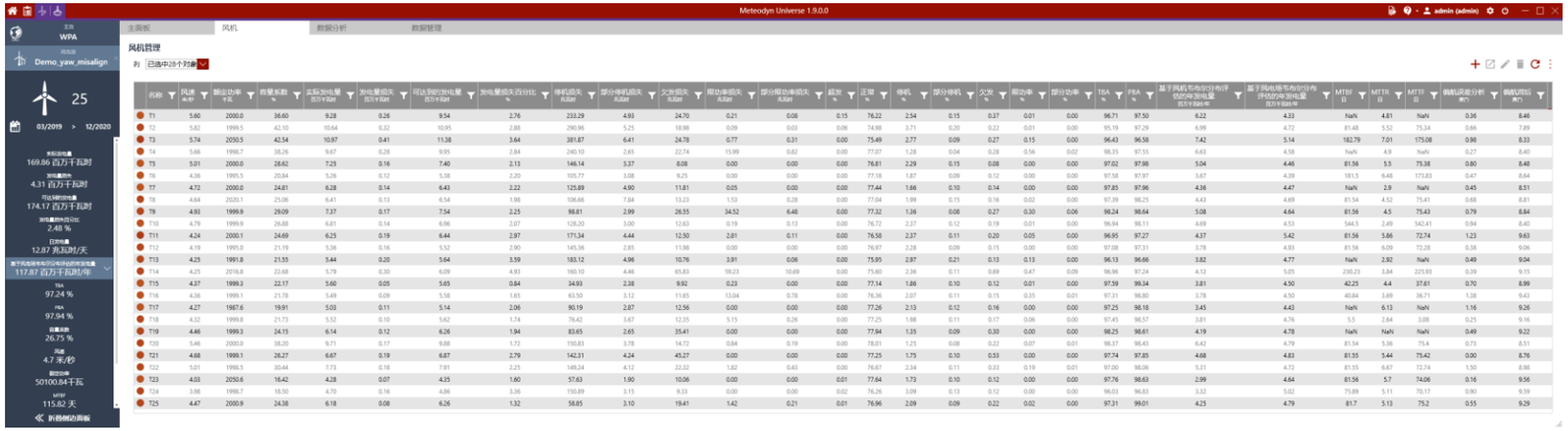
3 Performance analysis cases

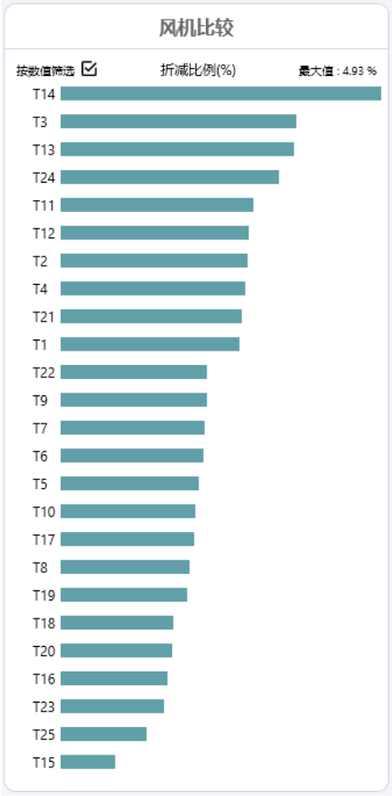
4 How to carry out wind farm performance analysis with WPA




# Identify the turbines that need to be analyzed

- Through the comparison and ranking of key operating indicators, the units that need to be analyzed can be quickly located
  - The largest loss of electricity generation
  - The highest proportion of anomalous data
  - TBA and PBA are the lowest
  - AEP performance indicators are the lowest
  - The failure indicators are the worst
  - Yaw error is maximum
  - .....



# Identify the turbines that need to be analyzed

- Through the comparison of the power curves of different units, the units that need to be analyzed can be quickly located

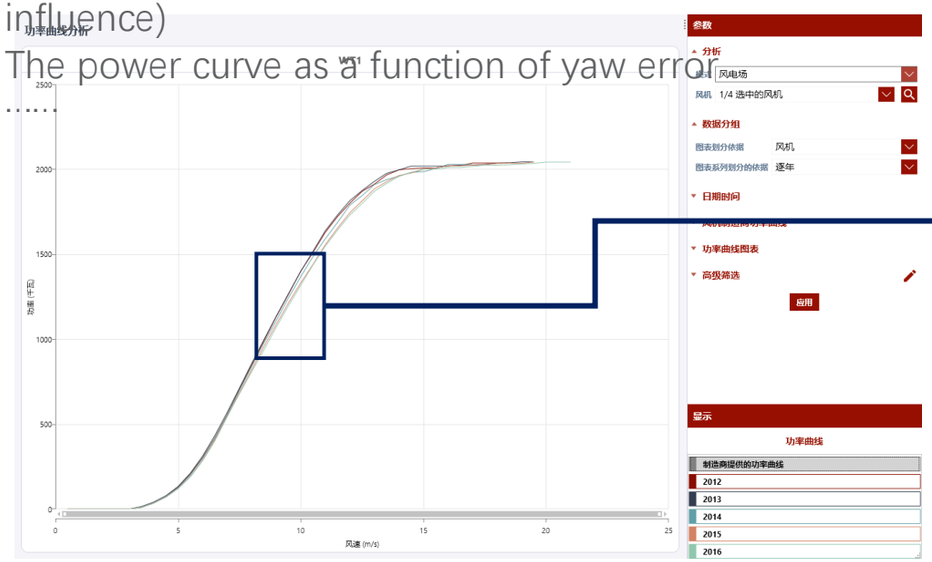
- Through the quantitative analysis of the proportions of different categories and the corresponding power generation losses, the main factors of power generation losses can be quickly identified and the analysis direction can be determined

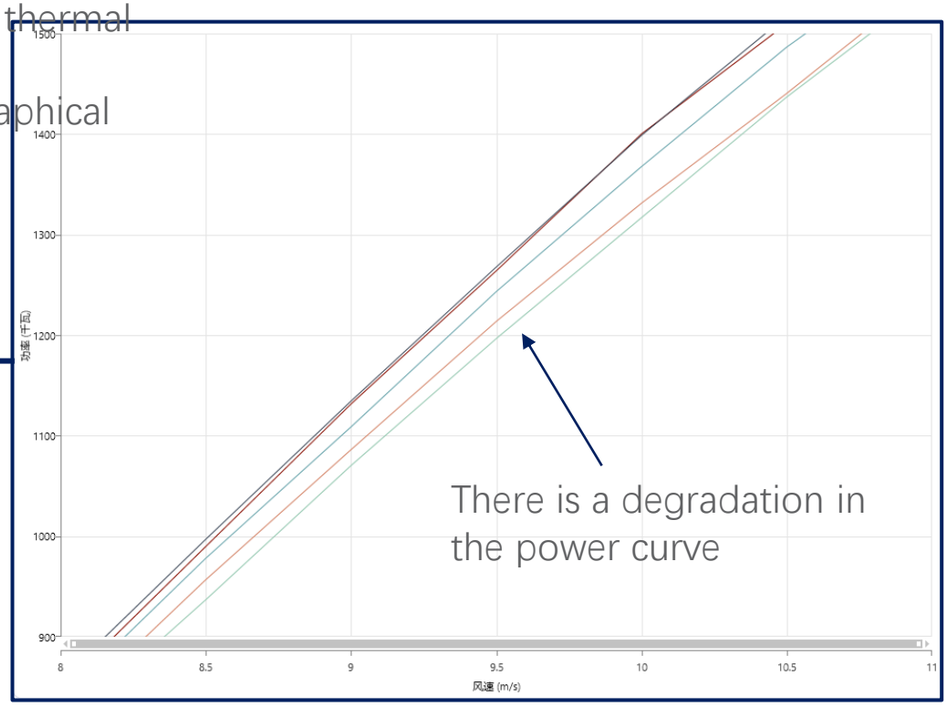





# Analyze the power curve from different dimensions

- By adjusting the analysis parameters, the influence of multiple factors on the power curve is studied
  - Interannual variation of the power curve (performance degradation)
  - Seasonal variation of the power curve (effect of changes in air density)
  - Diurnal variation of the power curve (influence of atmospheric thermal effect)
  - Relationship between power curve and wind direction (topographical influence)
  - The power curve as a function of yaw error
  - ……

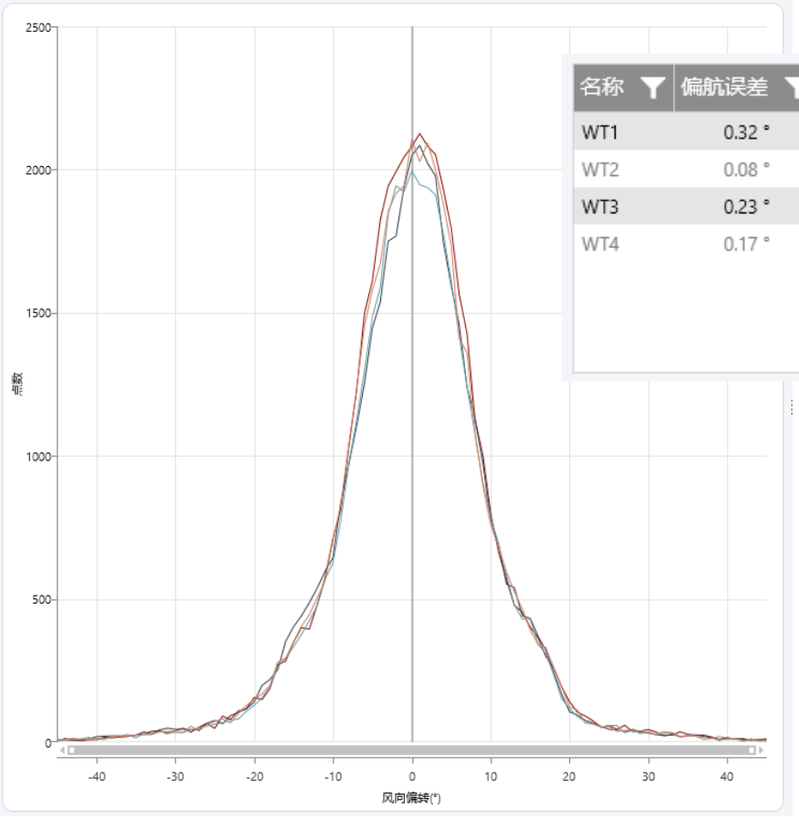


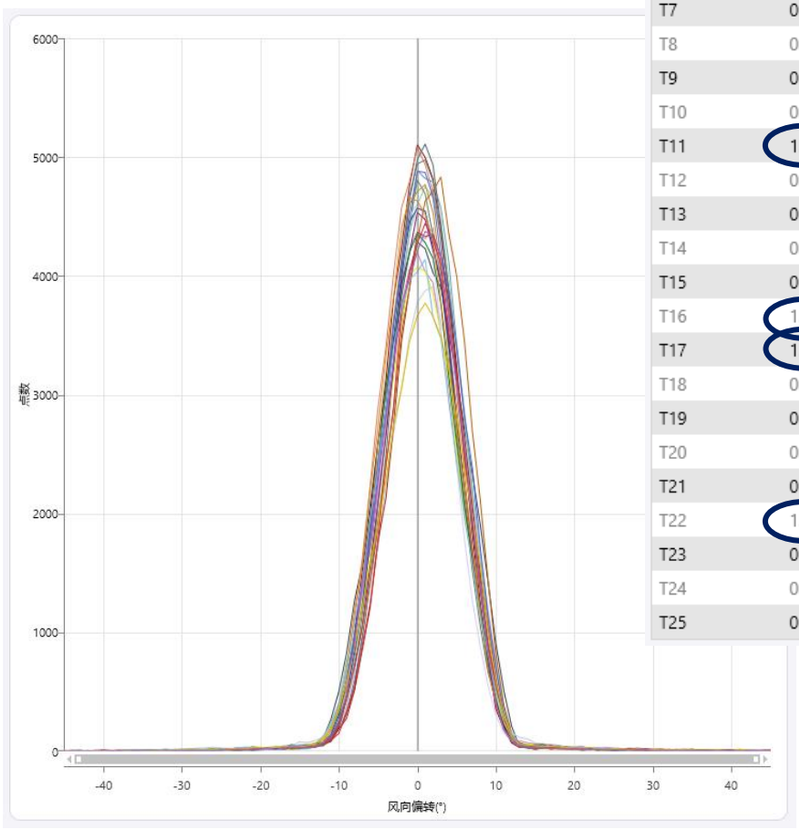
# Yaw error analysis

- The distribution of yaw error angles is given, and the average yaw error is evaluated

- Yaw lag angle is evaluated



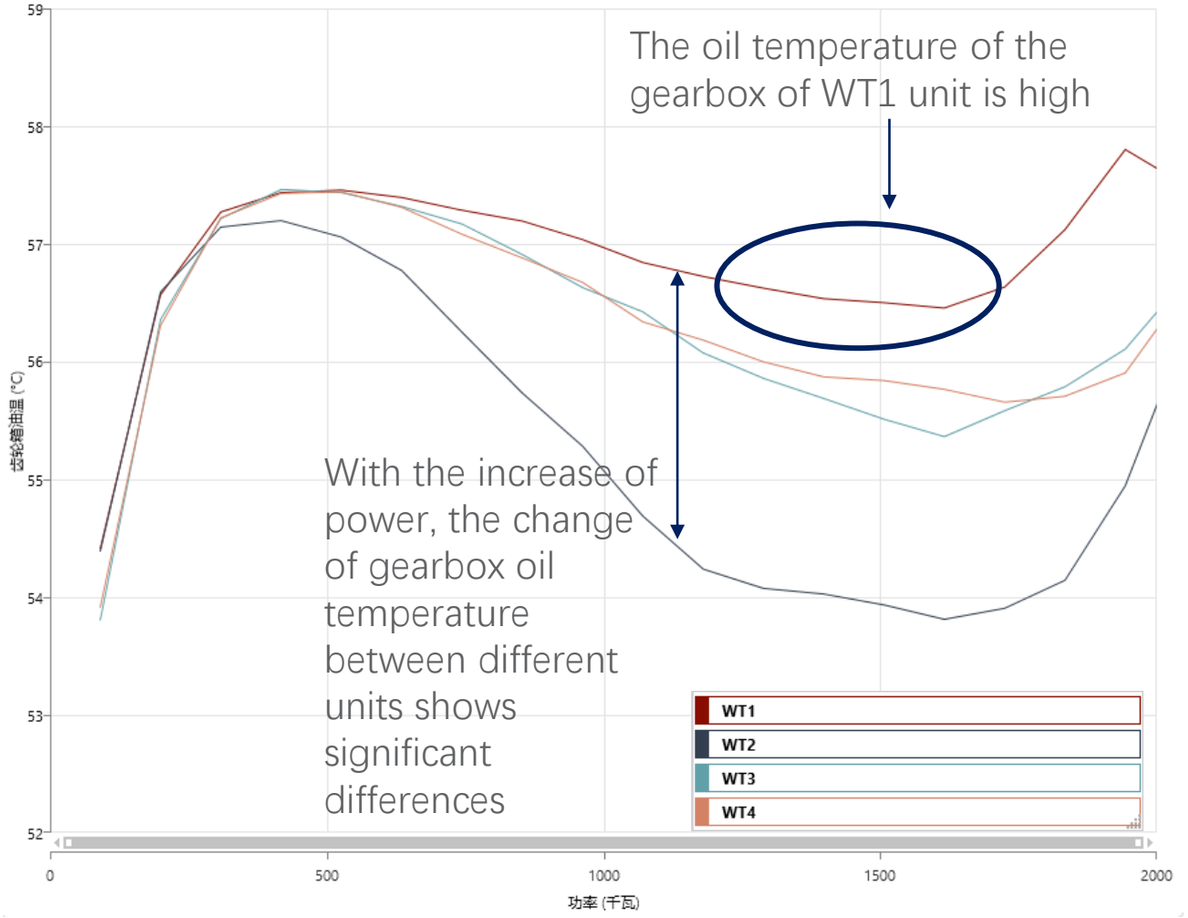

| 名称 | 偏航误差 | 偏航滞后 |
|---|---|---|
| WT1 | 0.32 ° | 10.16 ° |
| WT2 | 0.08 ° | 10.37 ° |
| WT3 | 0.23 ° | 10.12 ° |
| WT4 | 0.17 ° | 10.23 ° |




| 名称 | 偏航误差 | 偏航滞后 |
|---|---|---|
| T1 | 0.36 ° | 8.46 ° |
| T2 | 0.66 ° | 7.89 ° |
| T3 | 0.98 ° | 8.33 ° |
| T4 | 0.27 ° | 8.40 ° |
| T5 | 0.80 ° | 8.48 ° |
| T6 | 0.47 ° | 8.64 ° |
| T7 | 0.45 ° | 8.51 ° |
| T8 | 0.68 ° | 8.81 ° |
| T9 | 0.79 ° | 8.84 ° |
| T10 | 0.94 ° | 8.40 ° |
| T11 | 1.23 ° | 9.63 ° |
| T12 | 0.38 ° | 9.06 ° |
| T13 | 0.49 ° | 9.04 ° |
| T14 | 0.39 ° | 9.15 ° |
| T15 | 0.70 ° | 8.99 ° |
| T16 | 1.38 ° | 9.43 ° |
| T17 | 1.16 ° | 9.26 ° |
| T18 | 0.25 ° | 9.16 ° |
| T19 | 0.49 ° | 9.22 ° |
| T20 | 0.73 ° | 8.51 ° |
| T21 | 0.00 ° | 8.76 ° |
| T22 | 1.50 ° | 8.98 ° |
| T23 | 0.16 ° | 9.56 ° |
| T24 | 0.90 ° | 9.59 ° |
| T25 | 0.55 ° | 9.29 ° |

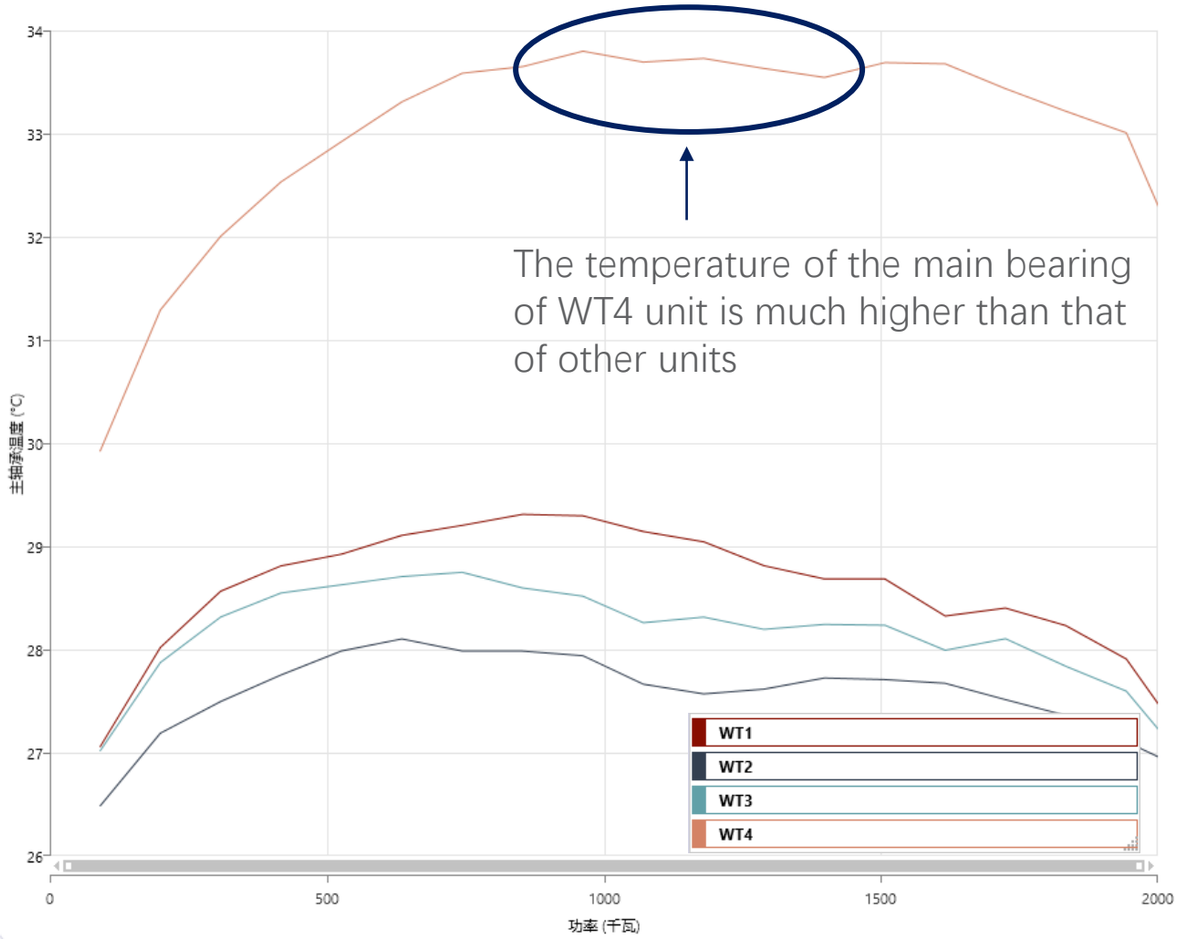
# Characteristic curves analysis

- Pre-configured for quick analysis
  - Air temperature - power
  - Cabin temperature - power
  - Stator temperature - power
  - Gearbox oil temperature - power
  - Main bearing temperature-power
  - Power - wind speed
  - Pitch - wind speed
  - Pitch - power
  - Shaft speed - wind speed
  - Shaft speed - power
- You can also freely select Y-axis and X-axis variables to generate characteristic curves
- A third variable can be selected to group the curves for easy comparison and revealing deeper data relationships



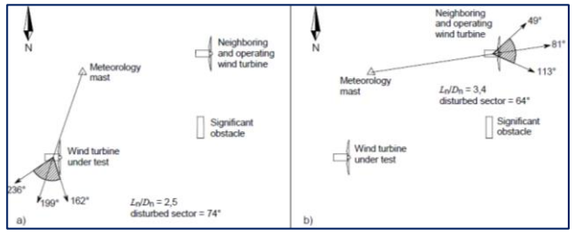
# Characteristic curves analysis

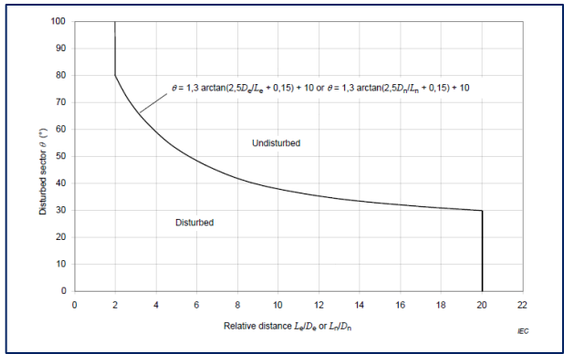
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  - Stator temperature - power
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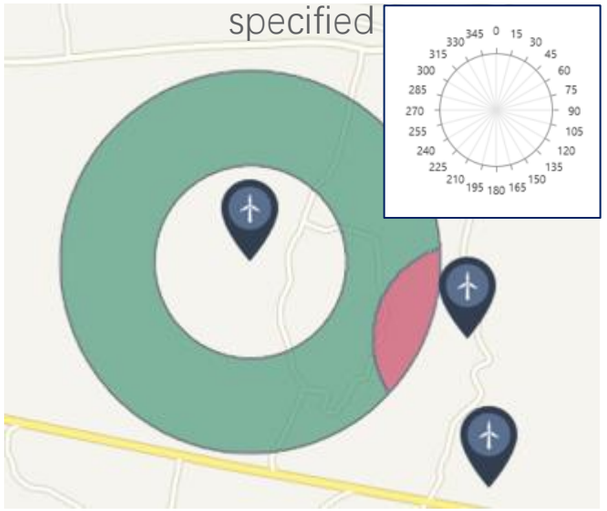
# IEC 61400-12-1-compliant mast location assessment

- Considering the IEC 61400-12-1 site selection requirements for masts used to measure the power characteristics of wind turbines and the corresponding exclusion sector requirements, combined with the dominant wind direction distribution in the proposed measurement period, the location suitable for the tower is recommended
- Calculate exclusion sector information for a specific mast location
- Automatically exclude the data of the corresponding sector in the NTF calculation

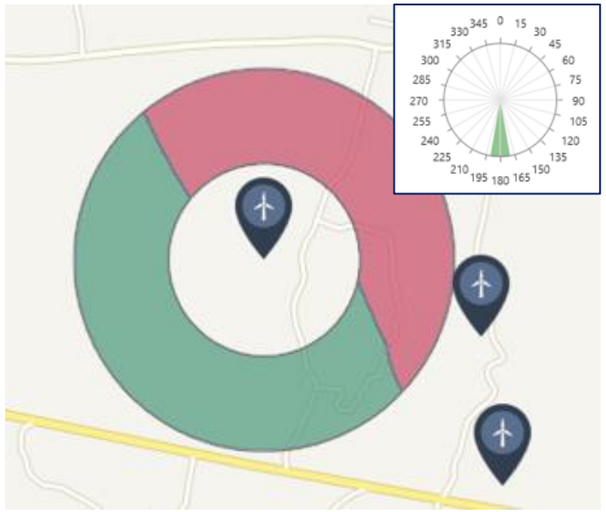


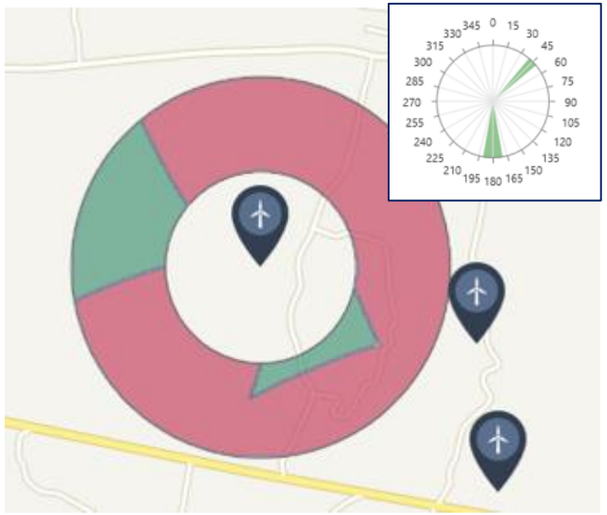

Measurement sector to keep: not specified



Keep one sector

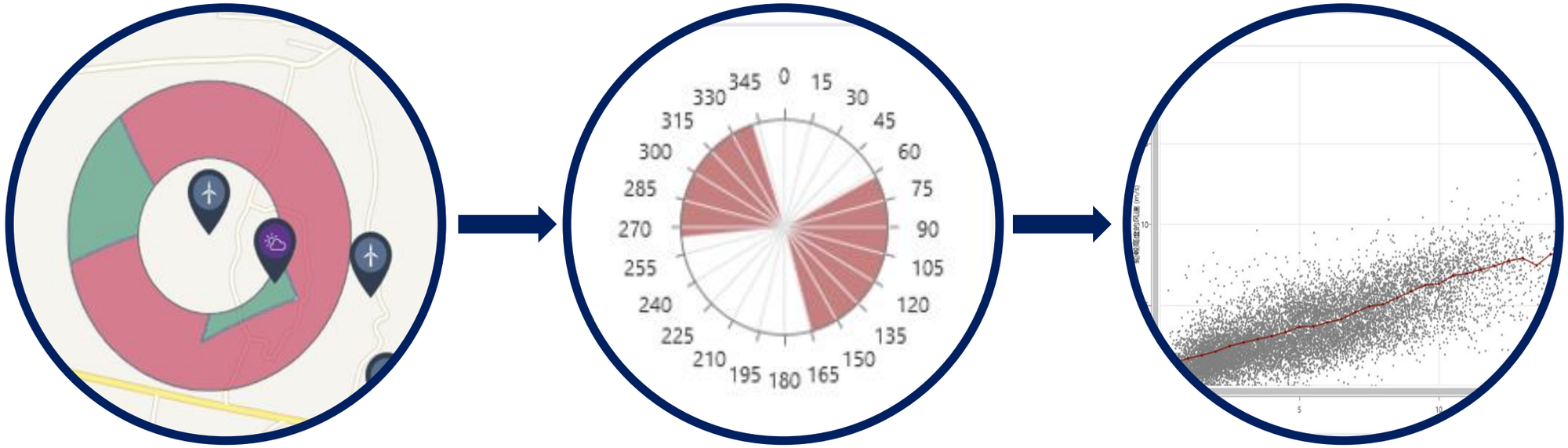


Keep two sectors



# IEC 61400-12-1-compliant mast location assessment

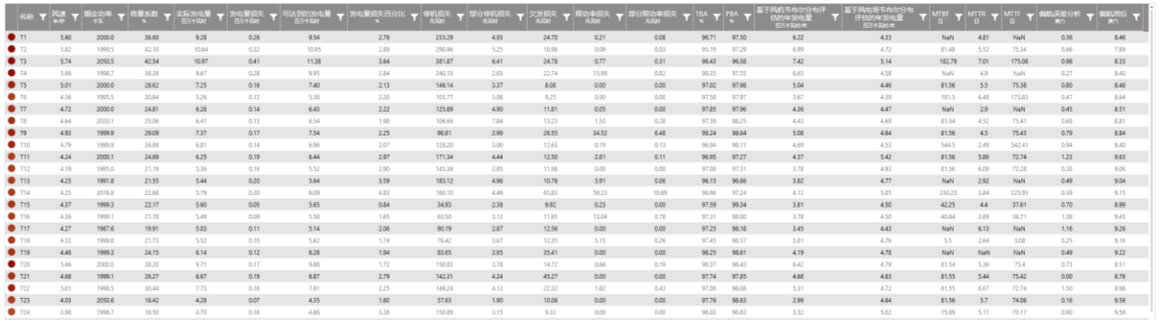
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- Automatically exclude the data of the corresponding sector in the NTF calculation



# Export data analysis results

có thể bảo họ export ra xong mình đọc từ đây ko?

- Export data analysis results as needed
  - Summary table of key metrics
  - Actual power curve
  - Characteristic curve data
  - A time series of operational status classifications
  - SCADA time series data in normal operation

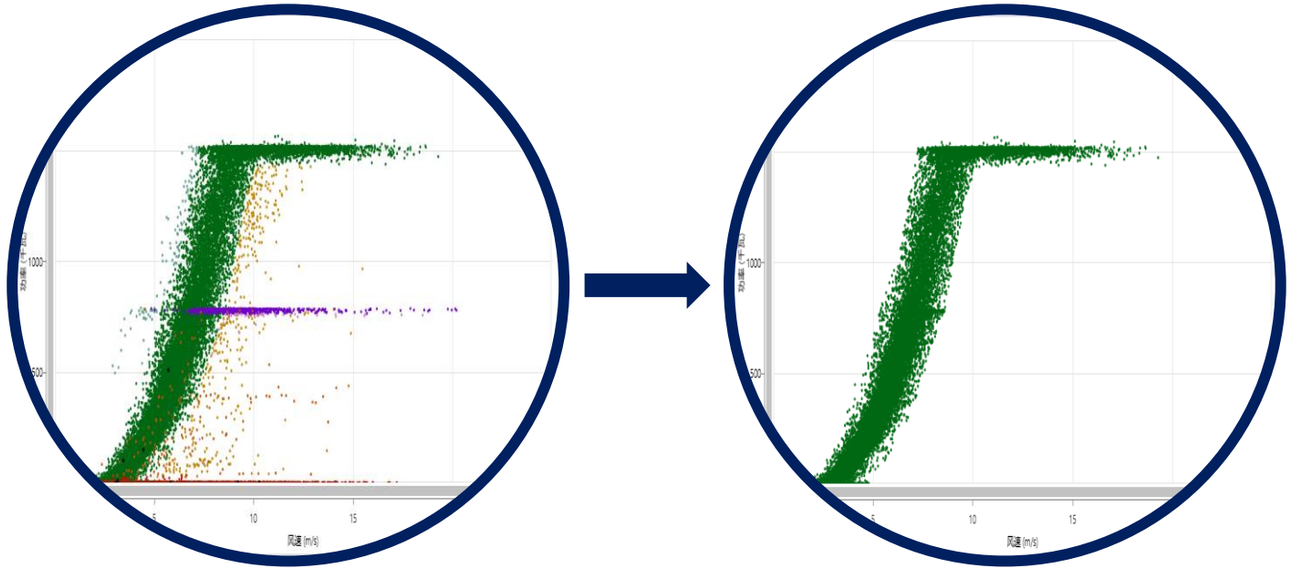


表格视图数据

输出到 CSV

| 风速 (m/s) | T1 | T2 | T3 | T4 | T5 | T6 | T7 | T8 | T9 |
|---|---|---|---|---|---|---|---|---|---|
| 0.5 m/s | 0 千瓦 | 0 千瓦 | 0 千瓦 | 0 千瓦 | 0 千瓦 | 0 千瓦 | 0 千瓦 | 0 千瓦 | 0 千瓦 |
| 1 m/s | 0 千瓦 | 0 千瓦 | 0 千瓦 | 0 千瓦 | 0 千瓦 | 0 千瓦 | 0 千瓦 | 0 千瓦 | 0 千瓦 |
| 1.5 m/s | 0 千瓦 | 0 千瓦 | 0 千瓦 | 0 千瓦 | 0 千瓦 | 0 千瓦 | 0 千瓦 | 0 千瓦 | 0 千瓦 |
| 2 m/s | 0 千瓦 | 1.43 千瓦 | 0 千瓦 | 0 千瓦 | 0 千瓦 | 0 千瓦 | 0 千瓦 | 0 千瓦 | 0 千瓦 |
| 2.5 m/s | 4.08 千瓦 | 22.87 千瓦 | 15.53 千瓦 | 5.59 千瓦 | 4.18 千瓦 | 13.79 千瓦 | 6.68 千瓦 | 10.3 千瓦 | 14.6 千瓦 |
| 3 m/s | 30.46 千瓦 | 70.93 千瓦 | 66.25 千瓦 | 38.32 千瓦 | 36.39 千瓦 | 56.42 千瓦 | 43.6 千瓦 | 48.45 千瓦 | 58.59 千瓦 |
| 3.5 m/s | 86.55 千瓦 | 145.1 千瓦 | 143.27 千瓦 | 103.61 千瓦 | 100.55 千瓦 | 122.86 千瓦 | 109.37 千瓦 | 117.46 千瓦 | 130.31 千瓦 |
| 4 m/s | 165.76 千瓦 | 222.83 千瓦 | 229.72 千瓦 | 182.09 千瓦 | 181.11 千瓦 | 198.45 千瓦 | 189.95 千瓦 | 198.43 千瓦 | 208.08 千瓦 |
| 4.5 m/s | 262.91 千瓦 | 310.39 千瓦 | 336.88 千瓦 | 279.32 千瓦 | 275.97 千瓦 | 283.64 千瓦 | 285.64 千瓦 | 298.36 千瓦 | 302.97 千瓦 |
| 5 m/s | 374.46 千瓦 | 421.61 千瓦 | 472.82 千瓦 | 395.06 千瓦 | 391.12 千瓦 | 387.41 千瓦 | 398.71 千瓦 | 420.2 千瓦 | 422.72 千瓦 |
| 5.5 m/s | 508.15 千瓦 | 561.98 千瓦 | 646.34 千瓦 | 541.9 千瓦 | 530.13 千瓦 | 520.87 千瓦 | 536.41 千瓦 | 575.66 千瓦 | 566.72 千瓦 |
| 6 m/s | 673.4 千瓦 | 737.93 千瓦 | 848.07 千瓦 | 727.36 千瓦 | 705.46 千瓦 | 684.81 千瓦 | 710.52 千瓦 | 762.46 千瓦 | 743.83 千瓦 |
| 6.5 m/s | 866.61 千瓦 | 941.05 千瓦 | 1082.35 千瓦 | 939.3 千瓦 | 905.38 千瓦 | 875.83 千瓦 | 906.97 千瓦 | 972.99 千瓦 | 936.66 千瓦 |
| 7 m/s | 1096.58 千瓦 | 1173.25 千瓦 | 1334.93 千瓦 | 1183.64 千瓦 | 1133.6 千瓦 | 1079.08 千瓦 | 1124.94 千瓦 | 1203.45 千瓦 | 1157.88 千瓦 |
| 7.5 m/s | 1332.13 千瓦 | 1406.3 千瓦 | 1576.74 千瓦 | 1426.75 千瓦 | 1363.59 千瓦 | 1296.83 千瓦 | 1350.36 千瓦 | 1420.82 千瓦 | 1390.35 千瓦 |
| 8 m/s | 1549.9 千瓦 | 1628.48 千瓦 | 1770.33 千瓦 | 1641.88 千瓦 | 1582.41 千瓦 | 1505.12 千瓦 | 1555 千瓦 | 1624.58 千瓦 | 1594.87 千瓦 |
| 8.5 m/s | 1742.32 千瓦 | 1803.84 千瓦 | 1908.62 千瓦 | 1811.13 千瓦 | 1753.49 千瓦 | 1679.45 千瓦 | 1725.95 千瓦 | 1780.34 千瓦 | 1761.39 千瓦 |
| 9 m/s | 1866.8 千瓦 | 1914.36 千瓦 | 1980.35 千瓦 | 1909.99 千瓦 | 1865.21 千瓦 | 1815.69 千瓦 | 1838.38 千瓦 | 1885.74 千瓦 | 1875.34 千瓦 |
| 9.5 m/s | 1945.19 千瓦 | 1963.81 千瓦 | 2025.1 千瓦 | 1956.89 千瓦 | 1936.71 千瓦 | 1897.68 千瓦 | 1904.57 千瓦 | 1953.01 千瓦 | 1939.94 千瓦 |
| 10 m/s | 1977.32 千瓦 | 1984.41 千瓦 | 2038.66 千瓦 | 1976.6 千瓦 | 1970.66 千瓦 | 1951.83 千瓦 | 1944.11 千瓦 | 1983.51 千瓦 | 1971.27 千瓦 |
| 10.5 m/s | 1988.07 千瓦 | 1991.24 千瓦 | 2044.47 千瓦 | 1985.63 千瓦 | 1986.6 千瓦 | 1975.4 千瓦 | 1968.52 千瓦 | 2000.91 千瓦 | 1987.13 千瓦 |

确定



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1 Performance analysis solutions for wind farms

2 WPA technical basis

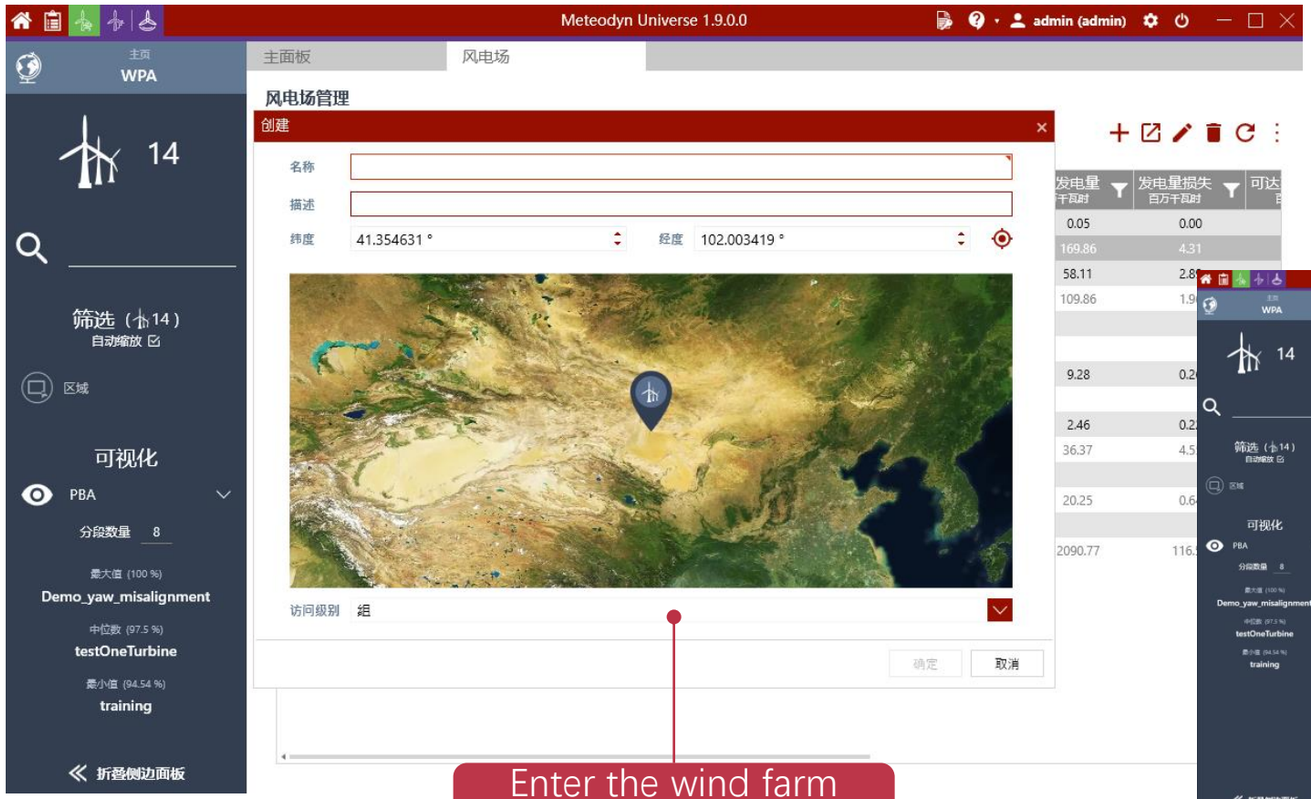
3 Performance analysis cases

4 How to carry out wind farm performance analysis with WPA



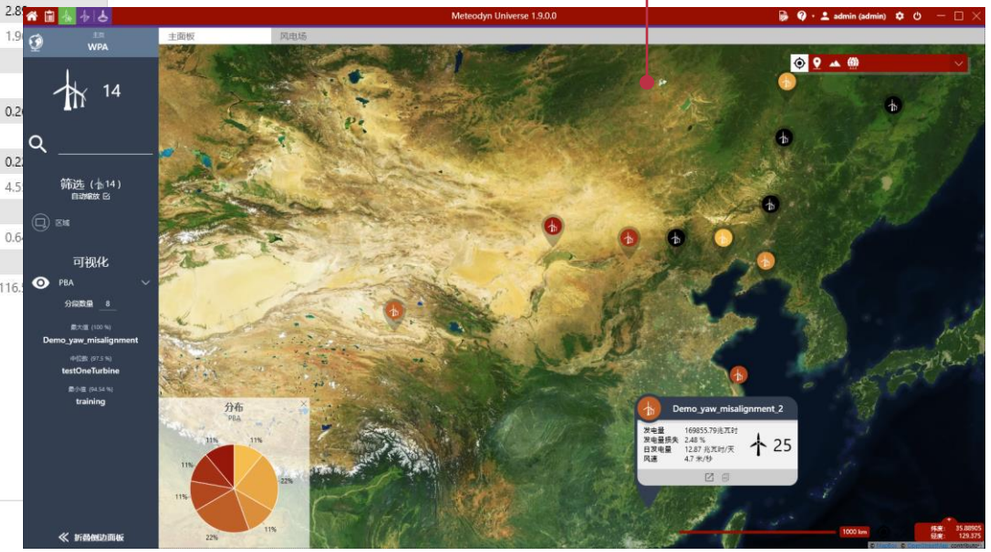
# WPA Operation: Create a wind farm

Create a wind farm | Data import and calculation | data analysis | Export the results



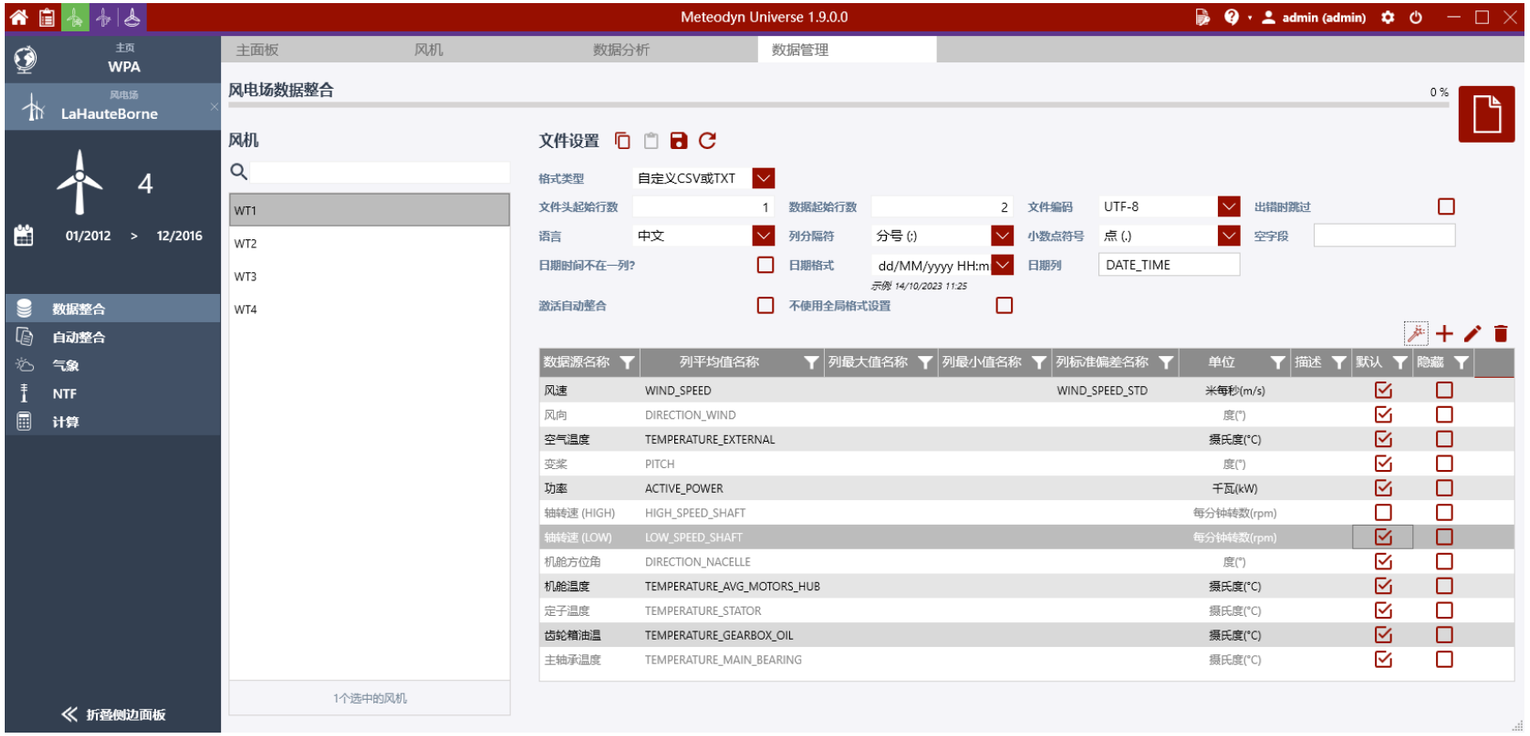

Enter the wind farm information

Map of the wind farms



# WPA Operation: Data Import

Create a wind farm → **Data import and calculation** → data analysis → Export the results



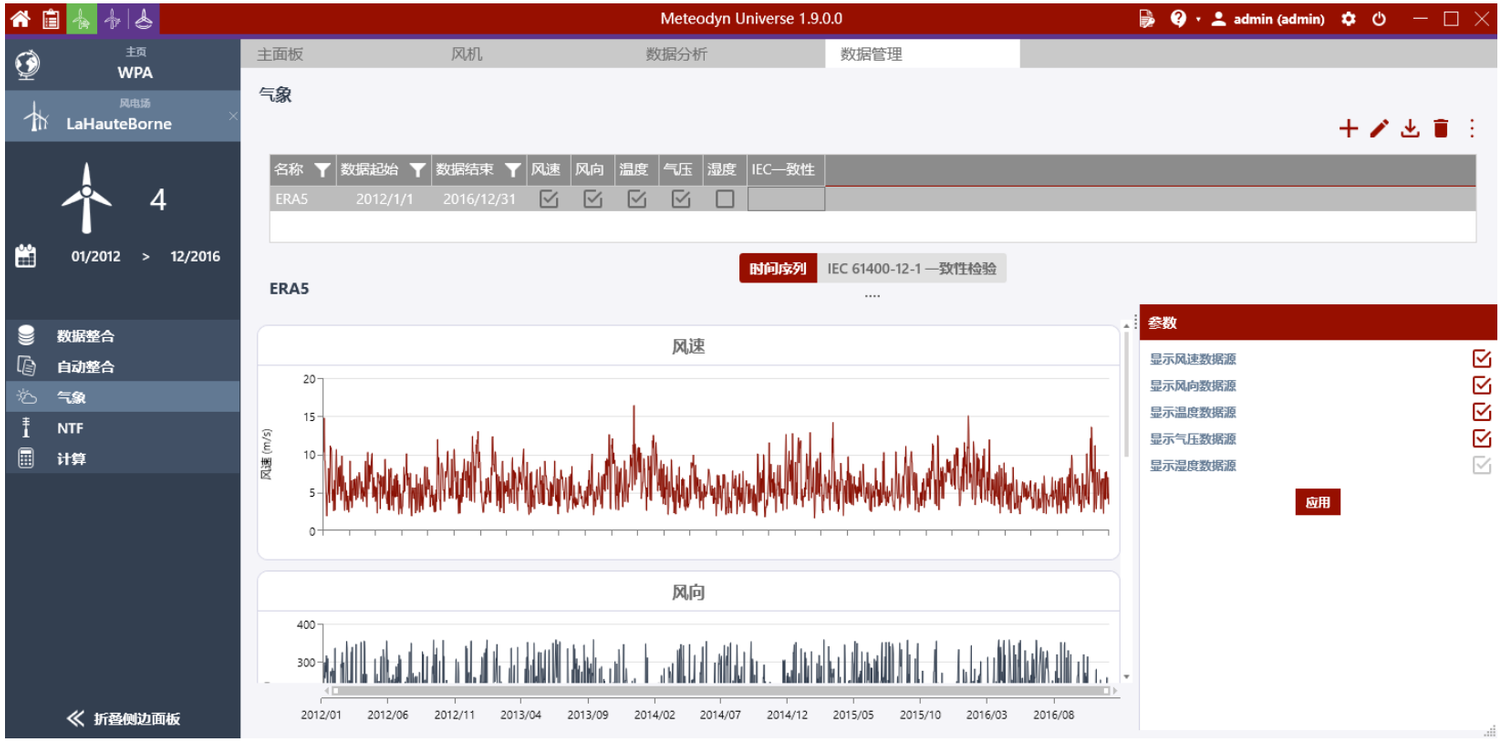

Import SCADA data

- Recommended time step: 10 minutes
- Variables that must be included
  - wind velocity
  - Active power
- Other variables that can be considered in the calculation (not mandatory)
  - Wind
  - Pitch
  - temperature
  - Shaft speed
  - Nacelle orientation
- Variables for more analysis (optional)
  - Nacelle temperature
  - Gearbox oil temperature

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# WPA Operation: Data Import

Create a wind farm → **Data import and calculation** → data analysis → Export the results

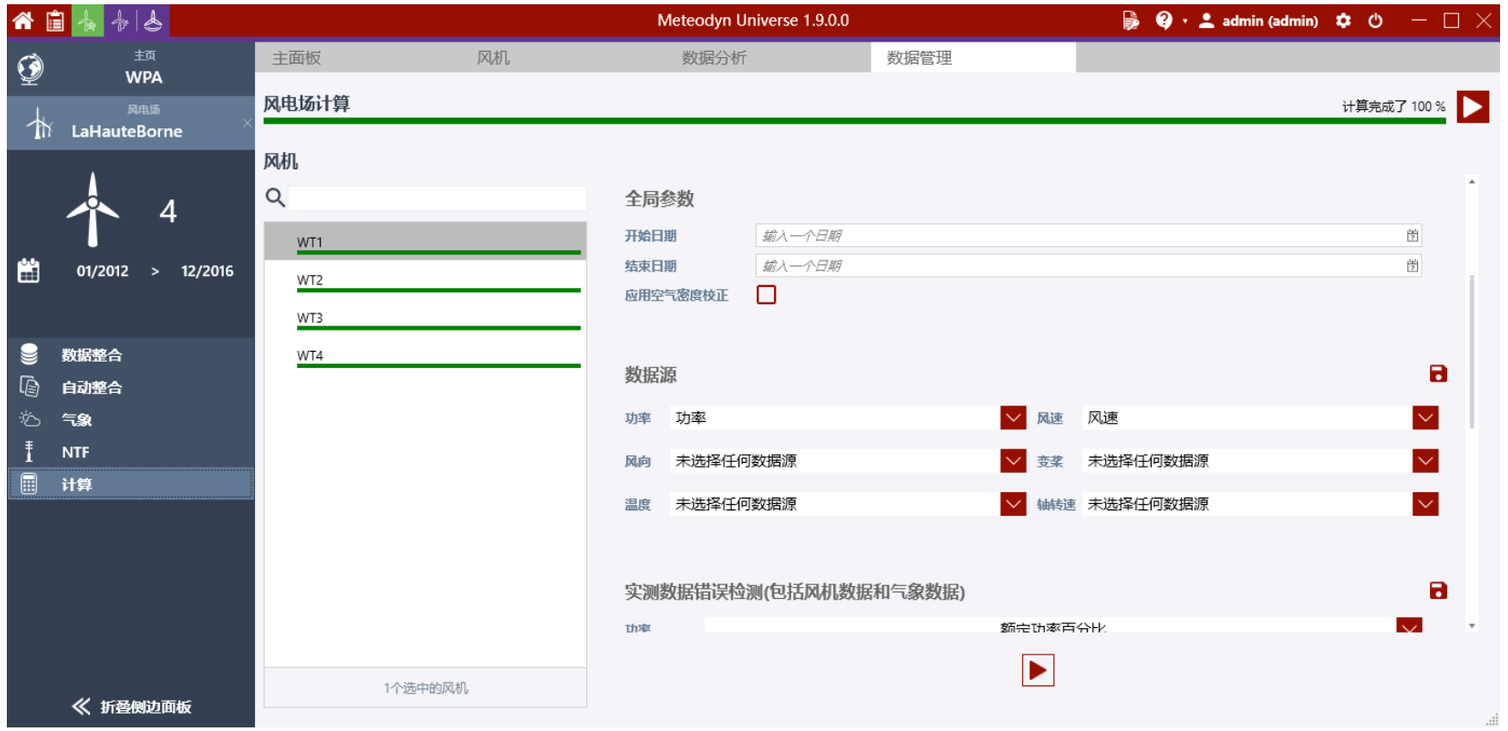



Import mast data (optional)

- Used for air density correction
- Used for NTF calculations

# WPA Operations: Data Calculations

Create a wind farm | Data import and calculation | data analysis | Export the results



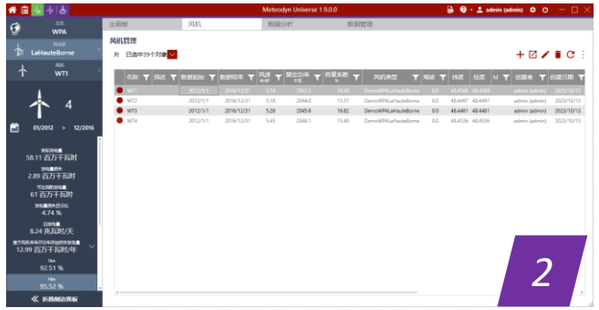

Set the calculation parameters

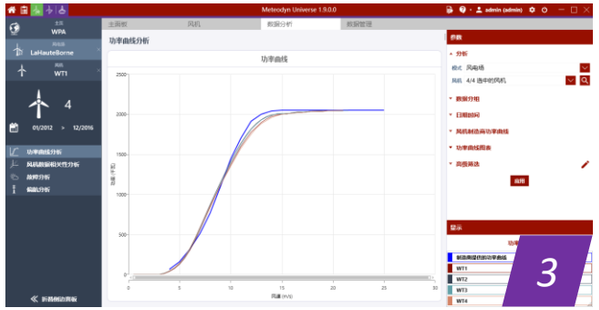
- Set the calculation time period
- Set the data source used for the calculation
- Adjust the invalid data detection rules
- Sets whether to enable air density correction

# WPA Operations: Wind farm-level analysis

Create a wind farm → Data import and calculation → **data analysis** → Export the results

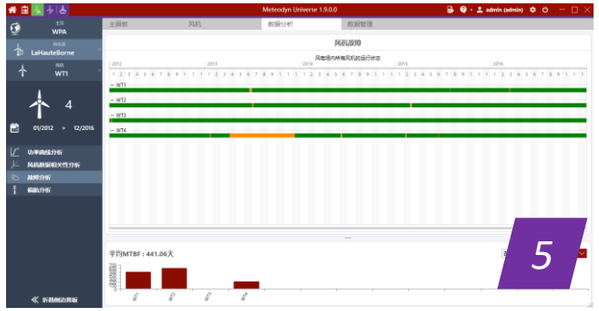


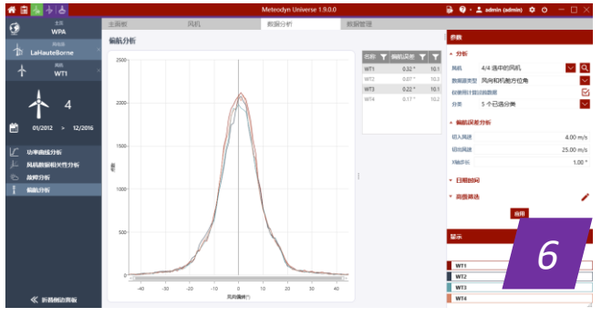










- The wind farm-level analysis focuses on the performance of all units and the comparison between the turbines

1. Overview of wind farm indicators
2. Performance index table of all units
3. Power curve analysis and comparison
4. Characteristic curve analysis and comparison
5. Failure analysis and comparison
6. Yaw analysis and comparison

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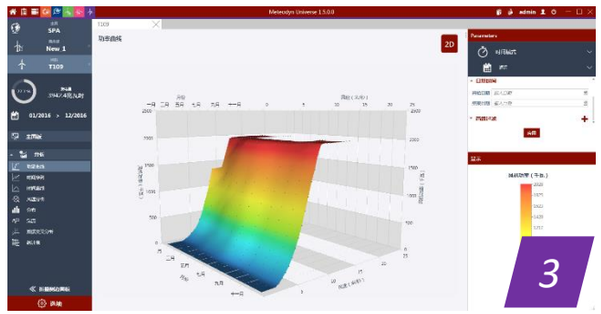
# WPA Operations: Wind turbine-level analysis

Create a wind farm | Data import and calculation | **data analysis** | Export the results



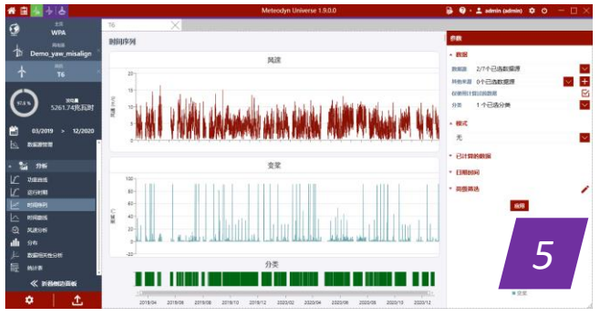


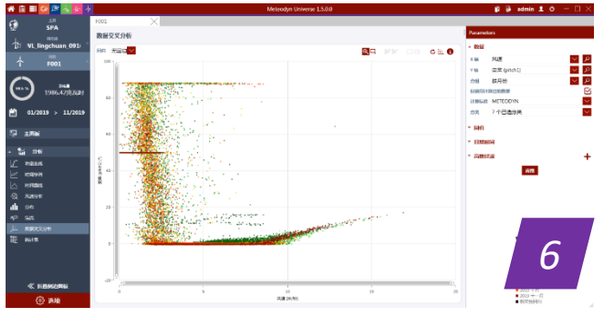










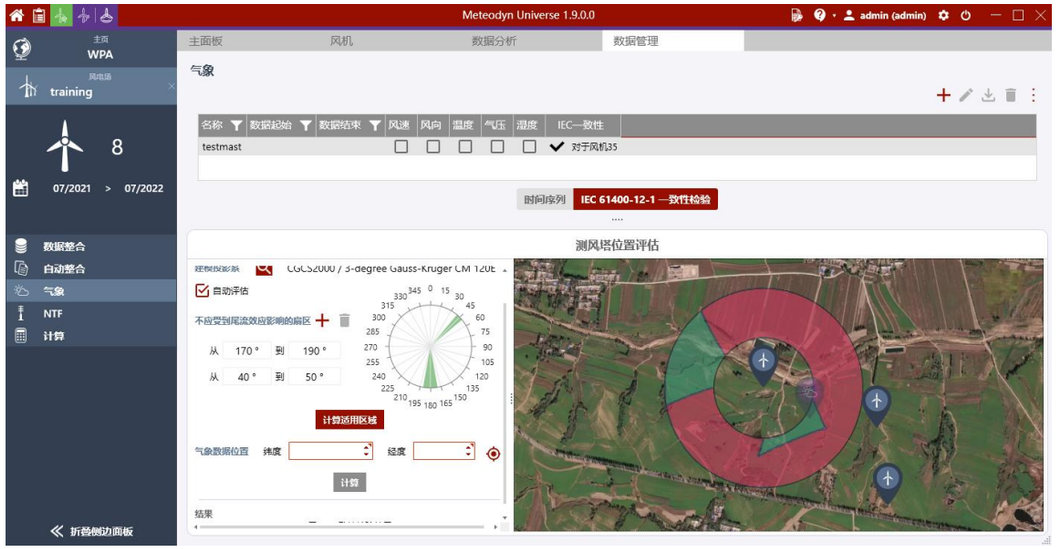

- Turbine-level analysis focuses on the detailed performance parameters of a specific unit and the study of specific problems

1. Overview of wind turbine indicators
2. Analysis of power curve/operating state classification
3. Analysis of the temporal variation of the power curve
4. Runtime analysis
5. Time series and time profile analysis
6. Cross-analysis of data

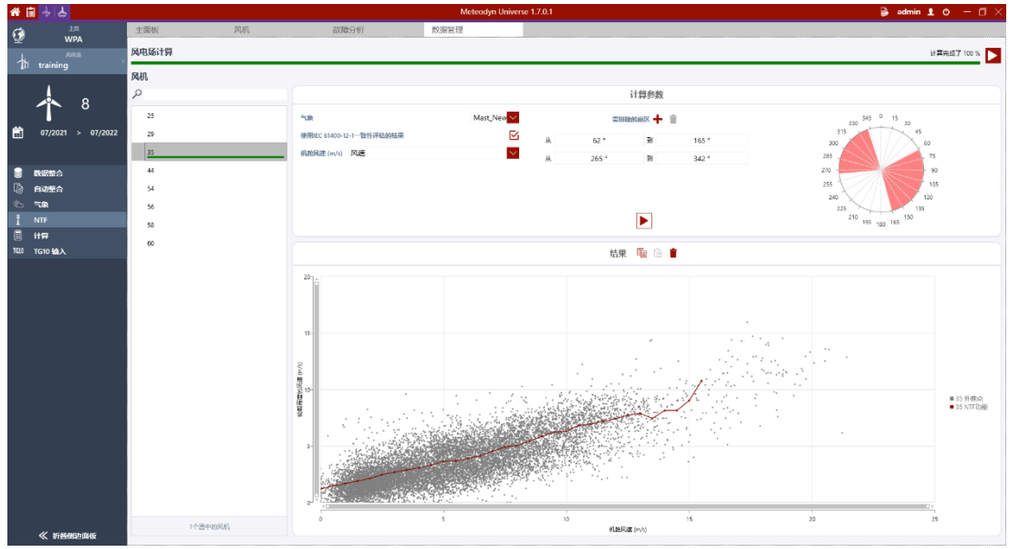
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# WPA Operations: Other analysis functions

Create a wind farm | Data import and calculation | **data analysis** | Export the results




IEC 61400-12-1 mast position assessment




NTF calculations

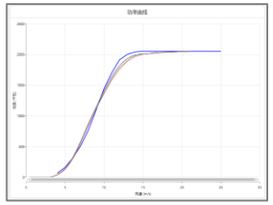


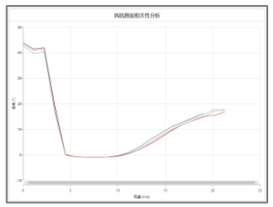
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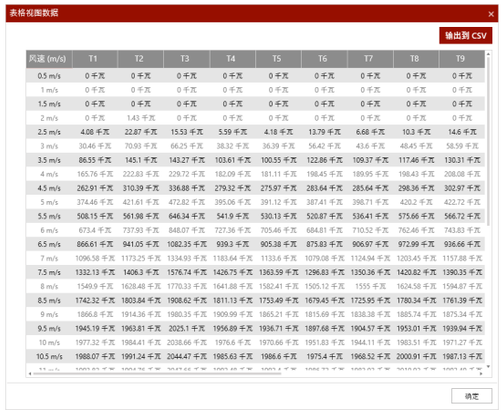
# WPA Operations: Data Calculations

Create a wind farm → Data import and calculation → data analysis → **Export the results**













- Export analyses in a variety of formats
  - Summary table of performance metrics
  - Power curve data
  - Data corresponding to characteristic curves and all other analysis curves
  - Time series data (filterable based on run status)
  - Statistical tables



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# Summary: The process of using WPA for performance assessment of wind farms

**SCADA data collection**

Collect SCADA data time series for all units in the wind farm with a time step of 10 minutes, including at least wind speed and power

**Data import and calculation**

Import SCADA data into WPA and perform automatic calculation, the software will automatically identify the operating status of the unit, and calculate the power curve and performance indicators

**Quick analysis**

Based on the main performance indicators, the energy efficiency of the wind farm is evaluated as a whole, and the direction of further analysis is clarified

**Export power curves and KPI**

Export power curves and performance indicator analysis results

**Perform other specific analyses**

Combined with the operation and performance indicators of the wind farm reflected in the rapid analysis step, a detailed analysis is carried out for specific problems

Wind farm performance assessment report

- Wind farm description
  - Basic information about the wind farm
  - Basic information of the wind turbine
  - SCADA Data Description
- Classification of the operating status of wind turbines
- Power curve analysis
  - Actual power curve
  - Analysis of factors influencing the power curve
  - Comparison of power curves of different units
- Performance indicator analysis
  - Analysis of actual power generation, required power generation and power generation loss
  - Availability analysis
  - IEC 61400-12-1 AEP Indicator Analysis
  - ……
- Other Analysis
  - Deviation Analysis
  - ……
- Specific analysis for turbines with poor performance

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# Example of a performance assessment report




METEODYN WPA



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01 The analysis of SCADA data is at the heart of the energy efficiency assessment of wind farms and is necessary for re-evaluation of in-service wind farms

02 WPA uses an efficient and robust method to identify and classify the operating status of wind turbines

03 WPA software can greatly improve the efficiency of SCADA data analysis, automatically calculate power curves and main energy efficiency indicators

04 Through KPI comparison, the turbines that need to be analyzed can be quickly determined

05 WPA provides problem-specific analysis capabilities that provide deeper insight into the causes of inefficient operation of some units



METEODYN WPA

# A powerful tool for performance assessment of wind farms

meteodyn
meteorology & dynamics

CLS
COLLECTE LOCALISATION SATELLITES



# Thank you

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